Project Overview and Description

Formula 1 Race Intelligence – Part 2 of 4

1. Project Summary

Formula 1 Race Intelligence is an interactive Tableau Public project built on 6,464 race records across 17 seasons. It turns race history into worksheets, dashboards, and a story that show driver performance, team championship trajectories, tyre strategy, Grand Prix host countries, podium achievements, and driver reliability.

2. Business Problem

Teams, broadcasters, and regulators all depend on race data, but the raw records are large and hard to read. Each group needs fast, visual answers to different questions. This project provides those answers in one interactive tool.

3. Objectives

- Benchmark team performance across seasons.
- Show how tyre compound and circuit type relate to results.
- Identify the strongest driver talent pools by nationality.
- Rank drivers and show podium achievements by driver and team.
- Show which countries host Grand Prix rounds.
- Show DNF patterns and judge whether reliability has improved.

4. Scenarios

Persona	Situation	Goal
F1 Team Performance Strategist	Senior strategist at a mid-tier constructor preparing for the next season	Benchmark points against Red Bull Racing and Mercedes, compare tyre strategies, and find talent pools for recruitment

Persona	Situation	Goal
Motorsport Broadcast Analyst	Data journalist preparing pre-race and post-race content	Pull the driver leaderboard, host countries, and podium distributions for broadcast graphics
FIA Technical Regulation Analyst	Compliance analyst studying retirements	Find whether failure types concentrate in particular teams or circuit types and whether reliability has improved

5. Business Requirements

- Interactive filters (season, team, driver, wet race, DNF reason).
- Clear comparison of Red Bull Racing, Mercedes, and a user-selected team.
- Dashboards that resize to different screens.
- A story that walks through the findings in order.
- Public access through Tableau Public.

6. Literature Survey

Motorsport analytics typically measures performance with points, podiums, wins, and retirement rates. Strategy studies focus on tyre compounds, pit stop timing, and circuit characteristics. Reliability studies examine the causes of retirements and how technical regulations change them. This project applies these measures in an interactive format. Add the specific sources you used here.

7. Social and Business Impact

- **Teams:** supports strategy, recruitment, and budget decisions.
- **Media:** speeds up the production of accurate graphics and statistics.
- **Regulators:** gives evidence for safety and technical rule discussions.
- **Fans:** makes the history of the sport easier to explore.

8. Tools and Technologies

Tool	Use
Tableau Public Desktop	Building worksheets, dashboards, and the story
Microsoft Excel	Source data (race_results, driver_nationality, circuits)
Tableau Public	Publishing and sharing
GitHub	Storing the workbook, data, screenshots, and documentation

9. Project Flow

1. Define the problem and scenarios.
2. Collect the data and connect it to Tableau.
3. Prepare the data and create calculated fields.
4. Build the worksheets.
5. Combine worksheets into dashboards.
6. Build the story.
7. Test performance.
8. Publish to Tableau Public.
9. Record the demo video and complete the documentation.

10. Deliverables

Deliverable	Description
Tableau workbook	F1_Race_Intelligence.twbx
Dashboards	4 dashboards
Story	Guided walkthrough of the findings
Public link	Tableau Public URL: <PASTE-LINK-HERE>
Demo video	Video link: <PASTE-LINK-HERE>

Deliverable	Description
Documentation	This four-part document set
Repository	GitHub repository with all project files